Section A

Answer **all** questions.

Write your answers in the spaces provided.

1. **Fig. 1.1** below shows an animal cell undergoing mitosis.

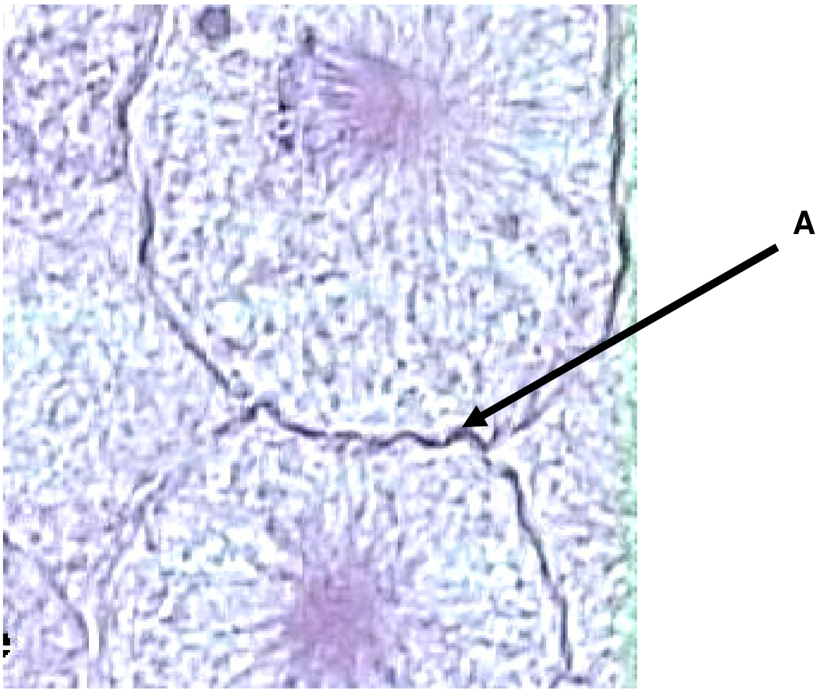

Fig. 1.1

(a) Identify this stage in the cell cycle and the structure labeled A as shown in **Fig. 1.1**.

Stage: ____________________ **A:** ____________________ [1]

(b) Suggest why this stage is the least frequently observed stage in mitosis.

____ [1]

Changes in DNA amount and chromosome numbers were monitored during one cycle of meiosis. Results are shown in **Fig. 1.2.**

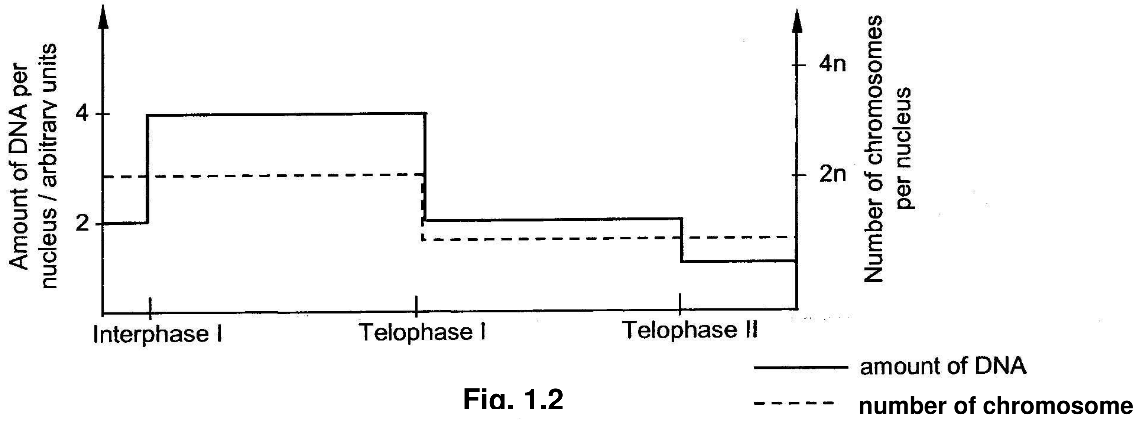

Fig. 1.2

(c) Explain why DNA amount and chromosome numbers remain constant during the two anaphases in **Fig. 1.2**.

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____ [2]

(d) List **two** ways in which prophase of *meiosis I* would differ from prophase in *mitosis*.

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____ [2]

(e) Explain how cancer is a result of uncontrolled cell division.

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____ [2]

Telomerase activity is generally activated in cancerous cells. Telomerase activity is detected in cancerous eukaryotic cells and germline cells. However, they are not present in prokaryotic cells such as bacteria.

(f) Explain why a telomerase inhibitor could form the basis of cancer chemotherapy.

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____ [2]

(g) Suggest why bacteria do not have telomerases.

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____ [2]

[Total: 12]

2. The following model in **Fig. 2.1** shows the initiation of translation in bacteria.

Fig. 2.1 shows the initiation of translation in prokaryotes.

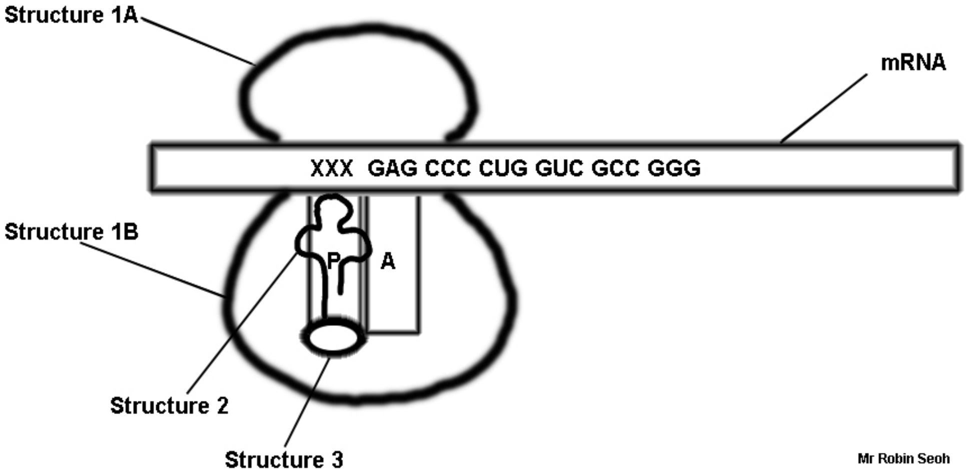

Fig. 2.1

(a) **(i)** Label the following structures

1A : ________________________________

1B : ________________________________

2 : ________________________________

3 : ________________________________ [2]

(ii) Describe the sequence in which the complex shown in **Fig. 2.1** is assembled. (Note: some of the structures are not shown)

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____ [3]

(iii) Describe the sequences following the formation of the complex shown in **Fig. 2.1**, leading to the formation of a <u>single</u> peptide bond.

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____ [3]

(b) **Fig. 2.2** shows the genetic code for the various amino acids (from www.sparknotes.com).

First position (5' end) / Third position (3' end)

	U	C	A	G	
U	UUU Phe UUC Phe UUA Leu UUG Leu	UCU Ser UCC Ser UCA Ser UCG Ser	UAU Tyr UAC Tyr UAA Stop UAG Stop	UGU Cys UGC Cys UGA Stop UGG Trp	U C A G
C	CUU Leu CUC Leu CUA Leu CUG Leu	CCU Pro CCC Pro CCA Pro CCG Pro	CAU His CAC His CAA Gln CAG Gln	CGU Arg CGC Arg CGA Arg CGG Arg	U C A G
A	AUU Ile AUC Ile AUA Ile AUG	ACU Thr ACC Thr ACA Thr ACG Thr	AAU Asn AAC Asn AAA Lys AAG Lys	AGU Ser AGC Ser AGA Arg AGG Arg	U C A G
G	GUU Val GUC Val GUA Val GUG Val	GCU Ala GCC Ala GCA Ala GCG Ala	GAU Asp GAC Asp GAA Glu GAG Glu	GGU Gly GGC Gly GGA Gly GGG Gly	U C A G

Amino acid names:

Ala = alanine	Gln = glutamine	Leu = leucine	Ser = serine
Arg = arginine	Glu = glutamate	Lys = lysine	Thr = threonine
Asn = asparagine	Gly = glycine	Met = methionine	Trp = tryptophan
Asp = aspartate	His = histidine	Phe = phenylalanine	Tyr = Tyrosine
Cys = cysteine	Ile = Isolevcine	Pro = proline	Val = valine

Fig. 2.2

Based on **Fig. 2.1** and **2.2**,

(i) Write down the sequence of the start codon represented by XXX in **Fig. 2.1**.

____ [1]

(ii) Write down the amino acid sequence of the polypeptide being formed.

____ [1]

(c) A mutation affecting a single base at the DNA level within the first seven codons causes the polypeptide formed to be truncated early, before the seventh amino acid is added to the polypeptide.

Write down the precise location of a nucleotide change in a specified codon (within the mRNA shown), along with the named type of mutation, that might lead to such a situation described.

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____ [1]

[Total: 11]

3. The *lac* operon (provided in **Fig. 3.1**) is a model used to describe how a group of genes involved in the metabolism of lactose are clustered together.

Fig. 3.1 shows the key features of the *lac* operon (from Wikipedia).

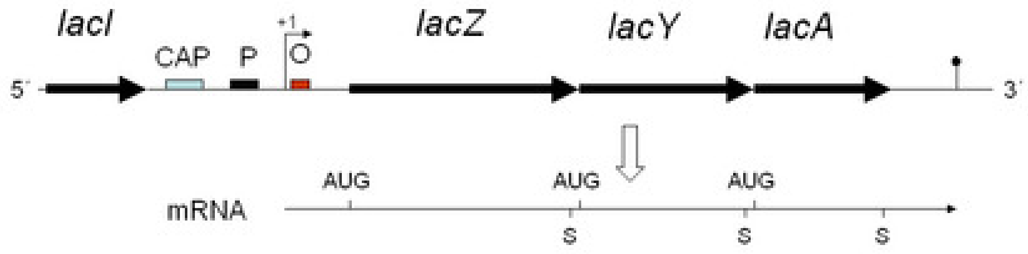

Fig. 3.1

(a) Explain what is meant by regulatory genes.

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____ [1]

(b) Name the three types of enzymes manufactured by each structural gene.

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____ [1]

(c) Explain with reference to **Fig. 3.1**, the control of transcription of the *lac* operon in the presence of lactose. (Reference to the CAP site is not required)

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____ [4]

(d) State clearly what AUG in the mRNA strand in **Fig. 3.1** represents.

____ [1]

A dual control of the *lac* operon exists, with the availability of glucose being a second control of transcription (refer to **Fig. 3.2**).

Fig. 3.2 shows the dual-control of the *lac* operon by the availability of both glucose and lactose (from Wikipedia).

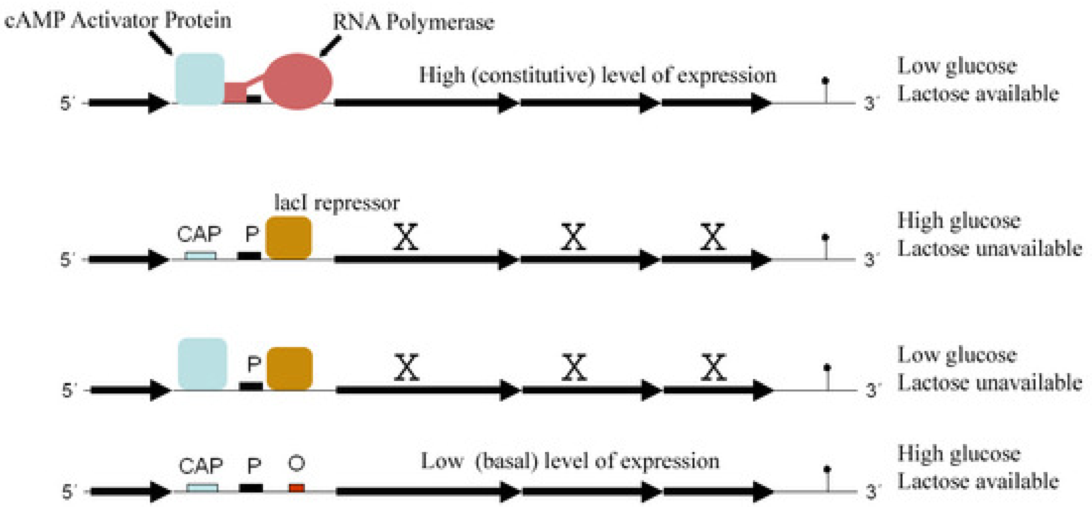

Fig. 3.2

(e) With reference to **Fig. 3.2**, explain the positive gene control of the *lac* operon.

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____ [4]

Fig. 3.3 below shows two eukaryotic models of transcription. **3.3a** shows a simple eukaryotic transcriptional unit while **3.3b** shows a complex eukaryotic transcriptional unit (from www.nature.com).

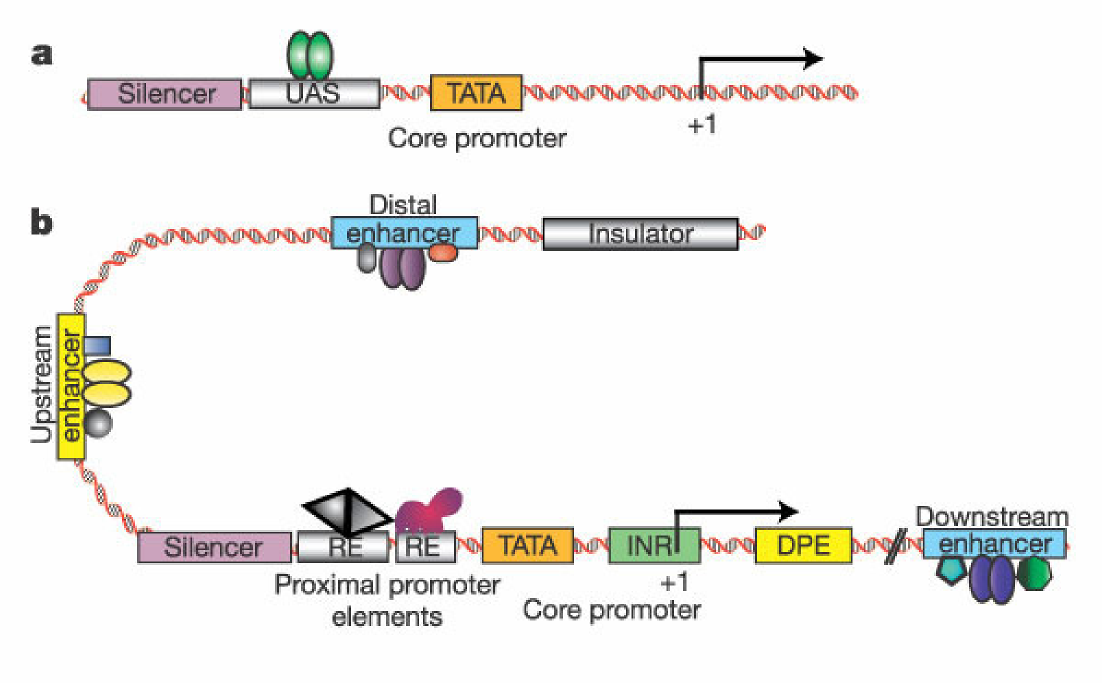

Fig. 3.3

(f) With the aid of **Fig. 3.3**, identify and explain the function of regulatory sequences, which are not found in the prokaryotic genome.

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____ [3]

[Total: 14]

4. Narcolepsy is a neurological condition characterized by excessive sleepiness in the day. The condition has been documented in humans and several mammals, including dogs. A pedigree of dogs carrying a form of the allele coding for narcolepsy, is shown in **Fig. 4.1.**

Fig. 4.1 shows the pedigree for inheritance of narcolepsy in dogs (modified from www.biomedcentral.com)

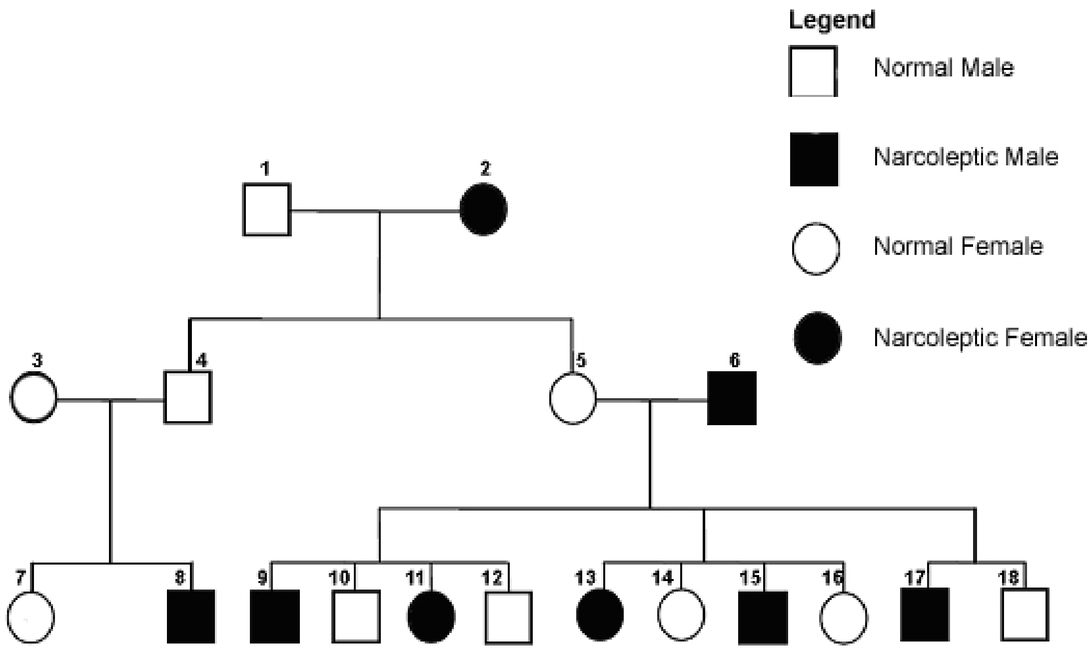

Fig. 4.1

(a) Given that the normal allele is denoted by **N** and the allele coding for narcolepsy is **n**, write down the genotypes of the following:

Individual 2 : _______________

Individual 4 : _______________

Individual 14 : _______________ [1]

(b) Using specific examples (individuals and their offspring) from the pedigree shown in **Fig. 4.1**, explain why the gene / allele coding for narcolepsy is autosomal and recessive.

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____ [2]

Molecular work on narcolepsy was focused on a neurotransmitter (neuropeptide) known as hypocretin, which naturally activates a G-protein coupled receptor. It was found that narcoleptic individuals either lacked hypocretin-producing cells in the hypothalamus, had a dysfunctional receptor, or did not possess cells able to receive hypocretin.

(c) Describe with elaborations two structural properties of a neuropeptide.

____ [2]

(d) Explain why all three types of molecular aberrations produce similar symptoms.

____ [3]

(e) Give two suggestions of potential methods for the treatment of narcolepsy.

____ [2]

[Total: 10]

5. **Fig. 5.1** shows the exchanges of substances between a mitochondrion and a chloroplast in a palisade mesophyll cell.

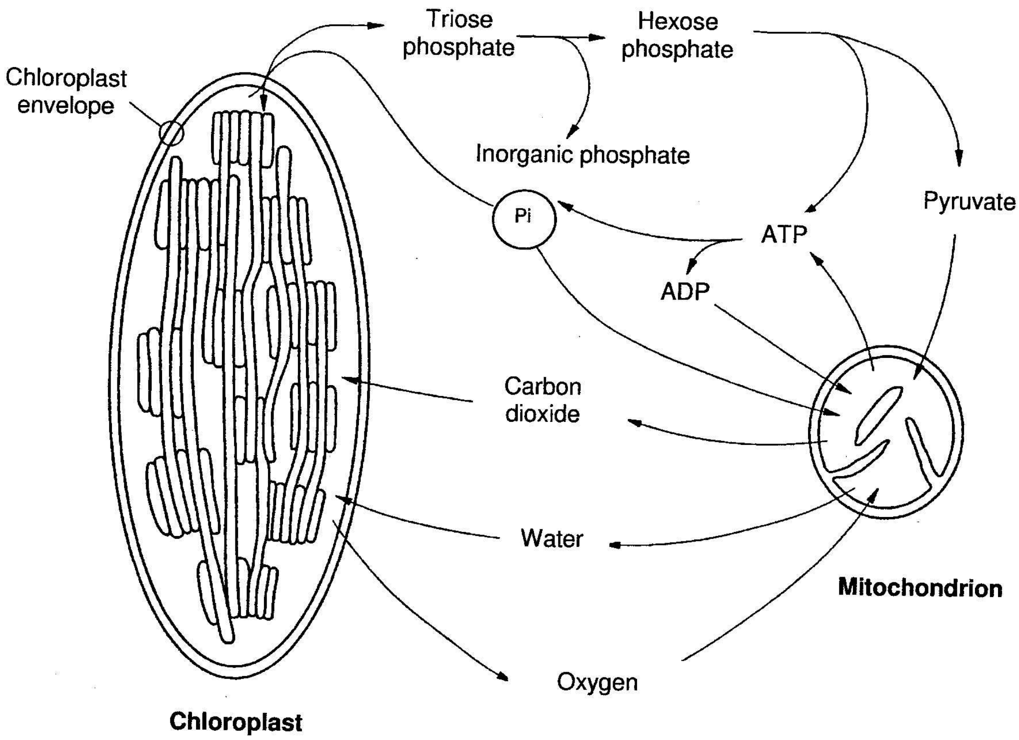

Fig. 5.1

A leafy shoot is sealed inside an air-tight transparent container. The concentration of oxygen in the atmosphere within this container can be measured. In the dark, the oxygen concentration falls. At high light intensities, the oxygen concentration increases. At a particular light intensity, the oxygen concentration in the container remains constant.

(a) With reference to **Fig. 5.1**, outline how the following links the metabolic pathways between the chloroplast and mitochondrion of this cell.

(i) triose phosphate and hexose phosphate;

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____ 2]

(ii) gaseous exchange.

[2]

(b) With reference to **Fig. 5.1**,

(i) explain how it is possible for the oxygen concentration to remain constant.

[2]

(ii) chloroplasts do not have the transporters that allow them to export ATP to the cytosol. How then does the rest of the cell get the ATP it needs for cellular metabolism?

[2]

In plants, the carbohydrates manufactured are channeled to form two types of macromolecules, namely cellulose and starch.

(c) State the two structural differences between cellulose and starch.

[2]

(d) Explain how these differences allow them to serve different functions in a plant cell.

[2]

[Total: 12]

6. **Table 6.1** was obtained from an experiment conducted on two liver cell cultures of rats, to study the expression of a particular type of glucose transporter (glucose transporter-4) on the cell membranes. The glucose level in the culture medium of both cell cultures was adjusted to 200 mg/dL, and a fixed volume of insulin was administered at time 0 to cell culture **B**, with the same volume of saline (weak salt solution containing 0.9% sodium chloride) added to cell culture **A** at the same time. At each the following times indicated in **Table 6.1**, a fixed volume of culture medium of both cultures was assessed for glucose concentration.

Table 6.1 shows the effect of insulin on glucose levels in culture media containing rat liver cell cultures.

Time (min.)	**Glucose level in culture A with saline added (mg/dL)**	**Glucose level in culture B with insulin added (mg/dL)**
0	200	200
10	192	182
20	184	151
30	174	115
40	163	93
50	153	89
60	149	92

Table 6.1

(a) Explain the purpose of setting up culture **A** when the aim of the experiment is to investigate the effect of insulin on the cell culture.

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____ [1]

(b) Describe the changes in glucose levels observed in both cell culture media.

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____ [2]

(c) Suggest an explanation for the glucose profile observed in the medium for cell culture **A** even though insulin is absent.

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____ [1]

The mechanism for glucose transport (mediated by insulin) was clarified and presented in **Fig. 6.2**. It has been established that the receptor for insulin is a tyrosine kinase receptor. The presence of insulin activates the receptor by inducing the formation of a homodimer from separate monomers located on the plasma membrane.

Fig. 6.2 shows insulin binding to its receptor **(1)**, which in turn starts many protein activation cascades **(2)**. These include: translocation of Glut-4 transporter to the plasma membrane and influx of glucose **(3)**, glycogen synthesis **(4)**, glycolysis **(5)** and fatty acid synthesis **(6)**. Enzymatic pathways are shown in the broken arrows while the fate of glucose was highlighted with bold arrows (from Wikipedia).

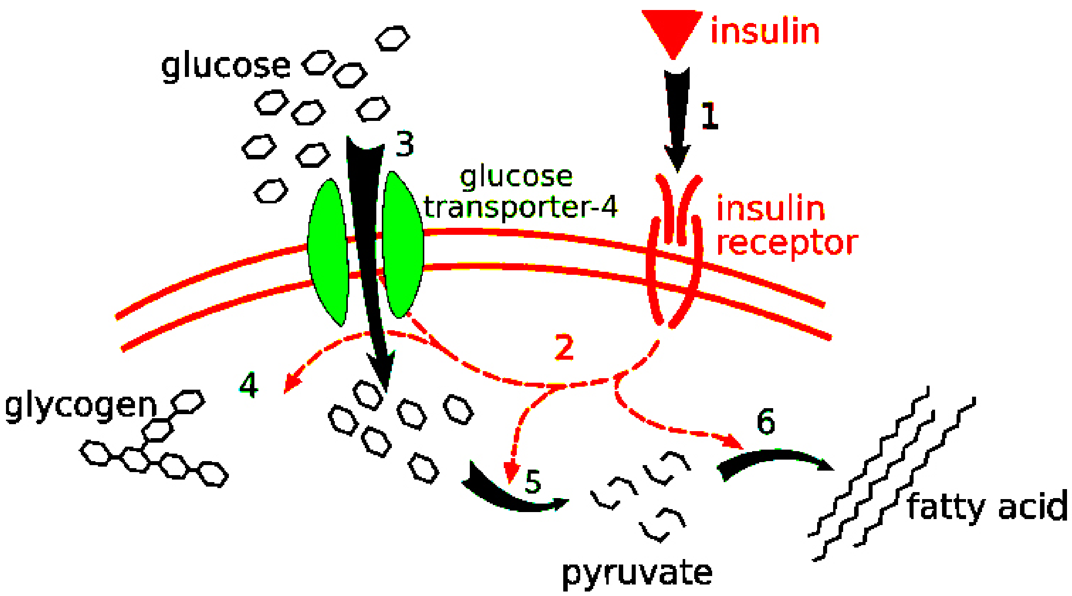

Fig. 5.1

(d) Describe the mechanism of glucose transport for glucose transporter-4.

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____ [1]

(e) Describe the mechanism of action of the enzymatic portion of the insulin receptor.

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____ [2]

(f) Suggest where glucose transporter-4 is normally located other than in the plasma membrane (before insulin binds to the receptor).

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____ [1]

(g) Explain the mechanism of cell signal amplification with the aid of **Fig. 6.2**.

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____ [2]

[Total: 10]

7. The Lake of Tanganyika in Africa has been considered by evolutionary scientists as a type of evolutionary playground for a type of fish known as the cichlid, in a way similar to how the Galapagos Islands allowed for adaptive radiation in the finches described by Darwin. In this lake, over 250 species of cichlids have evolved through speciation within a relatively short span of over 10,000 years. Other than the very diverse coloration and morphology seen in this myriad of species found, individual species also differ widely in feeding and mating behaviour.

For some species (termed as mouthbrooders), dozens of hatched young are kept safe in the enlarged mouths of parents until the fries are relatively advanced. There are also egg robbers that force the parents to release some of the young to feed on them. These are just two examples of unusual behaviour amongst many other types that exist in Lake Tanganyika

(a) Describe the process of adaptive radiation in the cichlids of Lake Tanganyika.

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____ [2]

(b) Explain how mouthbrooding behaviour in certain cichlid species might have evolved.

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____ [4]

A most peculiar behaviour is seen in the scale-eating cichlid, *Perissodus microlepis*, which feeds exclusively on the scales of other cichlids to survive. Morphs of this cichlid species are either right-mouthed or left-mouth. Scientists tracked the frequency of left-jawed individuals in the lake over a few years.

Fig. 7.1 below shows the frequency of left-jawed *Perissodus microlepis* in the lake of Tanganyika over a ten-year period.

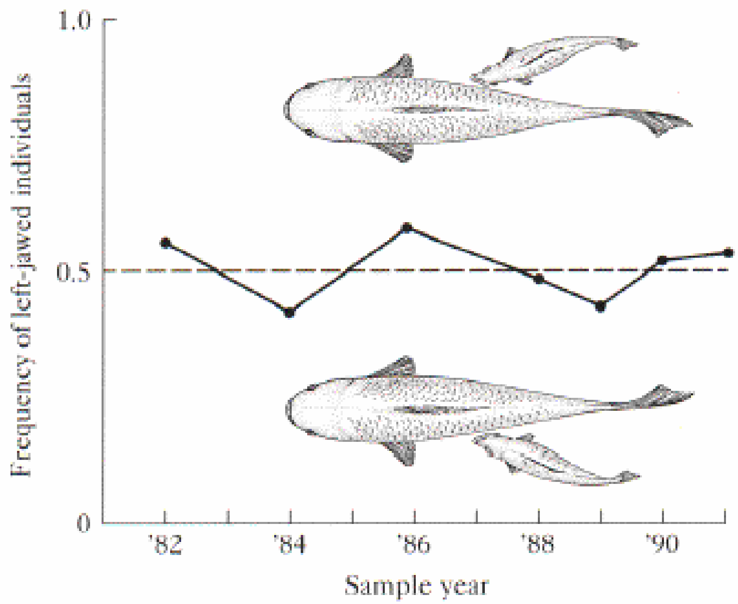

Fig. 7.1

(c) Account for the changes in frequency of the left-mouthed forms of *Perissodus microlepis* over the period of time surveyed, with reference to **Fig. 7.1**.

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____ [4]

[Total: 10]

Section B

Write your answers to this question on the separate answer paper provided.
Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a), (b)** etc., as indicated in the question.
Write down the question number you have attempted on the cover page.

8 **(a)** Compare the ways in which lambda phage and human immunodeficiency virus (HIV) reproduce themselves. [8]

(b) Describe the general structure of animal viruses and explain how these structures are adapted to infecting animal cells. [6]

(c) Outline how viruses are able to bypass the human defense mechanism to cause diseases. [6]

[Total: 20]

9 **(a)** Discuss how the membrane structure enables it to perform its function of being selectively permeable. [6]

(b) Explain how an organism is able to maintain a resting membrane resting potential of -70mV across the neurone membrane. [8]

(c) Describe the mechanism of synaptic transmission and contrast this with the transmission of a nerve impulse. [6]

[Total: 20]